

FINAL EXAM

Artificial Intelligence — Capstone Project

Student Guide — Design, Development & Defence

COURSE INSTRUCTOR

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1. GENERAL INSTRUCTIONS

The final exam consists of building a complete Artificial Intelligence solution that applies the skills acquired during the module. The work may be carried out individually, in pairs, in groups of three or in groups of four students.

The project must compulsorily include the following elements:

- A clearly defined AI task taken from one of the five families: classification, regression, object detection, clustering or recommendation systems
- A real or public dataset (Kaggle, UCI, Roboflow, etc.) with appropriate preprocessing and exploratory data analysis
- A complete pipeline: data → preprocessing → model training → evaluation
- Task-appropriate evaluation metrics (Accuracy/F1, RMSE/R², mAP/IoU, Silhouette, Precision@k, ...)
- A working demo: notebook, web app (Streamlit / Flask) or REST API
- Clean code organization and documentation (README + code comments)

Recommended stack: Python with scikit-learn, pandas and NumPy; TensorFlow/Keras or PyTorch for deep learning; YOLO / OpenCV for object detection; Matplotlib / Seaborn for visualization.

Work modalities

Team composition

The project may be carried out in one of the following configurations: individual work, pair (2 students), trio (3 students) or group of four (4 students).

The learning objectives and the expected delivery quality remain identical regardless of team size. A larger team is expected to demonstrate proportionally greater functional and technical ambition.

2. EVALUATION CRITERIA

The grading of the project rests on the five criteria below, assessed through the source code, the execution of the solution and the oral defence.

Criterion	Description
Problem & Data	Clear problem definition, relevant dataset, quality of EDA and preprocessing
Modeling	Appropriateness of the algorithm(s), feature engineering and tuning
Evaluation	Task-correct metrics, sound validation methodology, error analysis
Code & Reproducibility	Clean, organized code, README, reproducible end-to-end pipeline
Originality & Demo	Added value and a working demo (notebook, web app or API)

3. LIST OF PROPOSED TOPICS

Twenty topics are proposed below, covering the five AI task families. Each topic requires a full pipeline (data → model → evaluation) and a task-appropriate metric. The family is indicated in italics next to each topic.

Column A	Column B
01. Spam / phishing e-mail detection — <i>Classification</i>	11. PPE / safety-gear detection on site — <i>Object Detection</i>
02. Credit-card fraud detection — <i>Classification</i>	12. Retail product / fruit detection — <i>Object Detection</i>
03. Disease-risk prediction (e.g. diabetes) — <i>Classification</i>	13. Customer segmentation for marketing — <i>Clustering</i>
04. Product-review sentiment analysis (NLP) — <i>Classification</i>	14. News / document topic clustering — <i>Clustering</i>
05. House-price prediction — <i>Regression</i>	15. Network-traffic anomaly detection — <i>Clustering</i>
06. Hourly energy-consumption forecasting — <i>Regression</i>	16. Image colour segmentation (k-means) — <i>Clustering</i>
07. Used-car price estimation — <i>Regression</i>	17. Movie recommender (collaborative filtering) — <i>Recommendation</i>
08. Student-performance prediction — <i>Regression</i>	18. E-commerce product recommender — <i>Recommendation</i>
09. Face-mask detection (real-time) — <i>Object Detection</i>	19. Music / playlist recommendation — <i>Recommendation</i>
10. Vehicle & license-plate detection — <i>Object Detection</i>	20. Book recommender (content-based) — <i>Recommendation</i>

Note: any topic outside this list must be approved in advance by the instructor.

4. RECOMMENDED PROJECT STRUCTURE

The tree below illustrates the expected organization of the project. It reflects a clear separation between data, experimentation, source modules and the demo layer.

```
project/
├── data/
│   ├── raw/           # original datasets
│   └── processed/      # cleaned / feature data
├── notebooks/         # EDA & experiments (.ipynb)
├── src/
│   ├── data/          # loading & preprocessing
│   ├── models/         # training & inference
│   ├── evaluation/     # metrics & validation
│   └── app/            # demo (Streamlit / Flask / API)
├── models/            # saved weights / artefacts
├── requirements.txt    # dependencies
└── README.md          # documentation
```

5. SCHEDULE & DELIVERABLES

Key dates

Project submission date: the AI exam date provided by the administration.

Oral presentation date: *to be communicated later.*

Submission format

The deliverable is a single ZIP archive containing the following elements:

- The complete source code of the project
- The dataset, or a download link if it is large (Kaggle / Drive)
- The trained model / weights (or a script that reproduces them)
- A detailed report (PDF) describing the problem, methodology, results, plots and screenshots
- For large projects: upload to a personal Drive and place the link in a README.txt file included in the archive

6. METHODOLOGICAL ADVICE

To complete your project on time, you are strongly advised to follow the progressive approach below:

1. Define the problem and choose the right evaluation metric before writing any code.
2. Explore and clean your data (EDA, missing values, class balance, scaling/encoding).
3. Build a simple baseline first, then iterate toward stronger models.
4. Validate honestly: train/validation/test split or cross-validation; avoid data leakage and overfitting.
5. Track your experiments and version your work (Git recommended, with a clear commit history).

Good luck with the realization of your project.